4.0 Operational Procedures – CompTIA A+ 220-1202

4.1 Documentation and Support Systems

Ticketing Systems

- **Purpose:** Track and manage IT issues for accountability and resolution.
- **Key Components:**
  - User information
  - Device information
  - Issue description
  - Categories & severity
  - Escalation levels
  - Clear, concise communication
- **Documentation within Tickets:**
  - Issue description
  - Progress notes
  - Resolution

Asset Management

- Inventory lists
- Configuration management database (CMDB)
- Asset tags & IDs
- Procurement lifecycle
- Warranty and licensing
- Assigned users

Types of Documents

- Incident reports
- Standard Operating Procedures (SOPs)

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    - Software package custom installation procedure
- Onboarding/offboarding checklists
- Service-level agreements (SLAs) – internal and external
- Knowledge base/articles

4.2 Change Management Procedures

Documented Business Processes

- Rollback plan
- Backup plan
- Sandbox testing
- Responsible staff members

Change Management Process

- Request forms → Purpose → Scope → Change type:
    - Standard
    - Normal
    - Emergency
- Date/time of change → Consider change freeze & maintenance windows
- Affected systems / impact
- Risk analysis → Assign risk level
- Approvals → Change board review
- Implementation → Peer review → End-user acceptance

4.3 Workstation Backup and Recovery

Backup Methods

- Full backup
- Incremental backup

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- Differential backup
- Synthetic full backup

Recovery Methods

- In-place/overwrite
- Alternative location

Backup Testing & Rotation

- Test backups regularly
- Rotation schemes:
    - Onsite vs. offsite
    - Grandfather-Father-Son (GFS)
    - 3-2-1 rule (3 copies, 2 mediums, 1 offsite)

4.4 Safety Procedures

Electrical & Component Safety

- ESD straps & mats
- Equipment grounding
- Proper handling & storage of components
- Antistatic bags
- Cable management

Personal Safety

- Disconnect power before repairs
- Proper lifting techniques
- Fire safety
- Safety goggles & air filter masks

4.5 Environmental Impacts and Controls

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- **Material Safety Data Sheet (MSDS) compliance**
  - Battery disposal
  - Toner & device disposal
- **Temperature & Humidity Awareness**
  - Proper ventilation & dust cleanup
  - Compressed air/vacuums
- **Power Protection**
  - UPS (Uninterruptible Power Supply)
  - Surge suppressor

4.6 Privacy, Licensing, and Policy

- **Incident Response**
  - Chain of custody
  - Inform management/law enforcement
  - Preserve drive integrity
  - Document incident → Order of volatility
- **Licensing & DRM**
  - Valid licenses (perpetual, personal vs corporate, open-source)
- **Policies & Regulatory Compliance**
  - NDA / MNDA
  - Acceptable Use Policy (AUP)
  - Regulated data (PII, healthcare, credit card info)
  - Data retention requirements

4.7 Communication & Professionalism

- Professional appearance & attire (formal/business casual)
- Use proper language → Avoid jargon & slang
- Positive attitude, active listening, culturally sensitive
- Time management → Notify if late

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- Avoid distractions (phone/social media)
- Handle difficult customers → Clarify issues, remain professional
- Set expectations → Provide documentation & follow-up
- Protect customer confidential information

4.8 Basics of Scripting

- **File Types:** .bat, .ps1, .vbs, .sh, .js, .py
- **Use Cases:**
  - Automation of tasks
  - Restart machines, map drives, install apps
  - Automated backups & updates
  - Data gathering
- **Precautions:**
  - Avoid malware
  - Prevent system crashes
  - Do not unintentionally change system settings

4.9 Remote Access Technologies

- **Methods/Tools:**
  - RDP, VPN, VNC, SSH
  - RMM, SPICE, WinRM
  - Third-party tools: screen sharing, videoconferencing, file transfer, desktop management
- **Security Considerations:**
  - Encryption
  - Authentication
  - Permissions management

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4.10 Artificial Intelligence (AI) Basics

- **Application Integration:** Automation, analysis, productivity
- **Policy:** Appropriate use, plagiarism prevention
- **Limitations:** Bias, hallucinations, accuracy
- **Data Considerations:** Public vs. private data → security & privacy concerns

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